

TAFJORD, Norway

WMO: 012180

Lat: 62.2303N

Lon: 7.4217E

Elev: 56

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.3	19.8	-1.3	5.1	25.0	2.8	6.4	26.0	27.6	37.0	26.4	37.4	4.2	180	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	74.6	60.1	71.5	59.1	68.9	57.9	63.3	70.3	61.6	67.9	60.0	65.8	8.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	78.9	66.1	58.9	74.5	64.2	57.4	70.3	62.6	28.7	70.3	27.5	68.1	26.4	65.9	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.9	21.6	18.7	DB	14.4	80.7	3.2	5.0	12.1	84.3	10.3	87.3	8.5	90.1	6.2	93.7	(4)
(5)			WB	12.8	65.6	3.2	1.9	10.5	67.0	8.6	68.1	6.8	69.1	4.5	70.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	34.7	34.4	37.3	44.2	50.2	55.1	59.8	58.7	53.3	46.2	40.3	35.9
	DBStd	11.11	7.02	7.23	6.22	6.22	6.06	4.85	4.97	4.94	5.90	7.52	7.72	7.91
	HDD50	2512	473	437	393	192	70	9	1	1	30	165	301	440
	HDD65	6996	938	857	858	623	458	300	176	203	354	585	741	903
	CDD50	1021	0	0	1	19	77	162	304	272	128	46	10	2
	CDD65	29	0	0	0	0	0	3	14	9	2	1	0	0
	CDH74	126	0	0	0	0	3	12	68	40	2	1	0	0
	CDH80	14	0	0	0	0	0	0	11	3	0	0	0	0
Wind	WSAvg	6.5	9.6	8.8	8.2	6.5	5.8	4.3	3.6	3.9	4.6	7.0	8.3	7.9
Precipitation	PrecAvg	43.4	4.9	4.3	4.3	2.1	1.9	2.5	2.6	3.0	3.8	4.3	4.4	5.3
	PrecMax	60.2	14.0	8.8	18.2	7.6	4.3	6.7	4.3	6.6	9.6	9.9	12.2	10.9
	PrecMin	27.1	0.0	0.0	0.3	0.0	0.4	0.7	0.3	0.0	0.9	0.5	0.0	0.9
	PrecStd	8.2	3.7	2.5	3.9	1.5	0.9	1.3	1.0	1.4	2.4	2.2	3.5	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	54.3	56.0	65.3	73.1	75.6	80.5	79.0	73.2	69.4	62.4	57.5
		MCWB	42.8	43.4	44.1	49.9	56.1	59.6	63.4	60.3	57.8	55.0	50.1	46.2
	2%	DB	49.9	50.1	51.8	60.7	68.6	70.4	75.0	73.8	68.9	64.8	57.5	53.0
		MCWB	40.3	39.8	42.0	48.2	54.3	56.8	61.6	59.6	55.7	52.3	47.2	43.1
	5%	DB	47.4	47.1	49.1	57.8	64.6	66.9	71.3	70.6	65.9	60.4	54.3	50.0
		MCWB	39.2	38.1	40.1	46.5	52.4	55.4	60.6	59.3	54.3	48.7	44.4	40.7
	10%	DB	44.7	44.4	46.7	54.7	60.8	63.9	68.3	67.5	63.0	57.4	51.5	47.1
		MCWB	37.3	36.4	38.7	44.4	50.3	54.2	59.8	58.6	53.6	47.4	42.5	39.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	44.0	45.5	51.5	58.8	61.2	65.8	65.0	60.5	56.3	51.1	46.7
		MCDB	51.3	52.5	53.0	63.1	69.2	71.6	73.6	70.9	66.7	66.5	60.4	55.3
	2%	WB	41.3	41.3	43.2	49.0	55.5	58.5	63.9	63.0	58.3	53.2	47.8	44.2
		MCDB	48.2	47.5	49.1	58.8	65.2	66.6	71.4	69.1	64.9	62.0	55.4	51.8
	5%	WB	39.6	39.2	41.6	47.2	53.3	56.7	62.3	61.5	56.6	50.9	45.6	42.0
		MCDB	45.8	44.6	47.4	55.7	63.0	63.8	69.2	67.0	62.1	58.2	52.6	47.8
	10%	WB	38.2	37.7	39.8	45.6	51.2	55.4	60.5	60.0	55.1	48.8	43.6	40.1
		MCDB	43.4	42.9	45.3	53.0	59.1	61.9	66.4	65.0	60.3	55.2	49.7	45.3
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.7	8.9	11.5	13.2	12.6	11.7	11.3	11.0	9.3	7.7	8.1
		MCDBR	11.0	11.6	12.5	15.3	19.2	18.7	17.7	17.2	15.5	13.8	11.8	12.4
		MCWBR	5.8	5.9	7.0	7.9	8.9	8.6	8.0	7.0	6.7	6.4	6.0	6.9
	5% WB	MCDBR	10.2	10.8	11.6	13.6	17.9	15.6	16.0	13.1	13.2	12.7	11.6	11.5
		MCWBR	6.0	6.2	7.2	7.7	9.2	7.9	8.2	6.8	6.8	6.8	6.5	6.9
Clear-Sky Solar Irradiance	taub	0.233	0.285	0.321	0.352	0.356	0.356	0.368	0.358	0.327	0.302	0.240	0.208	
	taud	2.044	2.173	2.155	2.156	2.222	2.326	2.380	2.409	2.470	2.358	2.094	1.954	
	Ebn at Noon	139	211	245	262	272	275	268	260	248	207	133	83	
	Edh at Noon	11	21	31	38	39	36	33	30	23	17	10	6	
All-Sky Solar Radiation	RadAvg	52	204	554	1066	1439	1524	1443	1105	655	300	82	31	
	RadStd	11	49	98	137	128	154	128	111	93	47	20	4	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+2.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.94	+1.26	N/A	N/A	N/A	N/A	-247	-335	+106	N/A

Nomenclature: See separate page